Technical Report: Multi-Object Detection and Persistent ID Tracking

1. Model / Detector Used

YOLOv8s (You Only Look Once, version 8, small variant) from the Ultralytics framework was selected as the object detector.

YOLOv8 is a single-stage, anchor-free detector that processes an entire image in one forward pass, making it extremely fast while maintaining competitive accuracy. The "small" (`s`) variant was chosen as the optimal balance between inference speed and detection quality — important because the pipeline processes every frame of the video.

The model was used with its COCO-pretrained weights, filtering exclusively for **class 0 (person)**. A confidence threshold of **0.3** was applied to remove spurious low-confidence detections while retaining partially visible or distant players.

2. Tracking Algorithm Used

BoT-SORT (Bag of Tricks for Simple Online and Realtime Tracking with a Deep Re-identification Model) was selected as the multi-object tracker.

BoT-SORT extends the ByteTrack algorithm with two critical enhancements:

1. **Camera Motion Compensation (CMC):** Uses background feature matching (sparse optical flow or ECC alignment) to estimate the camera's ego-motion between frames. This compensates the Kalman filter predictions so that the tracker doesn't "lose" stationary or slow-moving subjects when the camera pans rapidly — a common scenario in sports broadcasts.

2. **Re-Identification (ReID) Module:** Extracts appearance feature embeddings for each detected subject and uses cosine similarity to re-associate tracks after occlusion. This is what allows the tracker to correctly re-assign the *same* ID to a player who was temporarily hidden.

3. Why This Combination Was Selected

Challenge	How YOLO + BoT-SORT Addresses It
Multiple overlapping players	YOLO detects each instance independently; BoT-SORT uses IoU + appearance matching to distinguish them
Camera panning / zooming	BoT-SORT's CMC compensates for camera ego-motion in the Kalman filter
Occlusion recovery	BoT-SORT's ReID features re-associate tracks after temporary disappearance
Fast motion	YOLOv8's per-frame detection + Kalman prediction handles rapid acceleration
Real-time feasibility	Single-stage detector + efficient tracker keeps processing fast

This combination was specifically chosen over simpler alternatives (e.g., DeepSORT, ByteTrack) because sports footage inherently involves aggressive camera work, making CMC a non-negotiable requirement.

4. How ID Consistency Is Maintained

The ID consistency mechanism works through a multi-stage association pipeline:

1. **Kalman Filter Prediction:** For each existing track, the Kalman filter predicts where the subject should be in the next frame (compensated by CMC for camera motion).

2. **First Association (High-confidence detections):** High-confidence YOLO detections are matched to predicted track positions using a combination of IoU (Intersection over Union) distance and appearance (ReID) cosine distance.

3. **Second Association (Low-confidence detections):** Remaining unmatched tracks are tentatively matched to lower-confidence detections using IoU alone — this recovers partially visible subjects.

4. **Track Buffer:** Unmatched tracks are not immediately deleted. They are kept in a "lost" buffer for a configurable number of frames. If a matching detection appears within that window, the original ID is restored.

5. **New Track Initialization:** Only completely unmatched detections that persist for a few consecutive frames are promoted to new tracks with new IDs.

5. Challenges Faced

- **ID Switches During Close Interactions:** When two players in similar jerseys run past each other in close proximity, the tracker occasionally swaps their IDs. This is a fundamental limitation of 2D bounding-box tracking.
- **Distant / Small Players:** Players far from the camera produce tiny bounding boxes near the detection threshold, causing intermittent detection losses.
- **Referee / Non-Player Detection:** The pipeline detects *all* persons, including referees, coaches, and ball boys. Additional classification logic would be needed to filter these out.

6. Failure Cases Observed

- **Prolonged Full Occlusion (>2 seconds):** When a player is completely hidden behind another for an extended period, the track buffer expires and a new ID is assigned upon reappearance.
- **Post-Goal Celebrations / Huddles:** When many players cluster tightly together, the detector may merge bounding boxes or the tracker may lose individual identities.
- **Rapid Camera Cuts:** Hard cuts (scene transitions) cause the CMC to produce invalid motion estimates for 1-2 frames, potentially disrupting all tracks.

7. Possible Improvements

1. **Fine-tuned Detection Model:** Train or fine-tune YOLOv8 on a sports-specific dataset to improve detection of small/distant players and to distinguish players from referees.

2. **Team Clustering:** Use jersey color histograms or a lightweight classifier to assign team labels alongside IDs.
3. **Top-Down Projection:** Apply homography transformation to project player positions onto a 2D pitch map for tactical analysis.
4. **Trajectory Smoothing:** Apply a post-processing smoothing filter to reduce jitter in bounding box positions.
5. **Ensemble Tracking:** Combine BoT-SORT output with an offline global re-identification pass to correct ID switches retroactively.
6. **3D Pose Estimation:** Integrate pose estimation (e.g., ViTPose) to improve re-identification using body keypoints rather than just appearance.